A3-PS036-TC161

PQC DNSSEC MTL Benchmark Using a Reference Implementation (PS-036)

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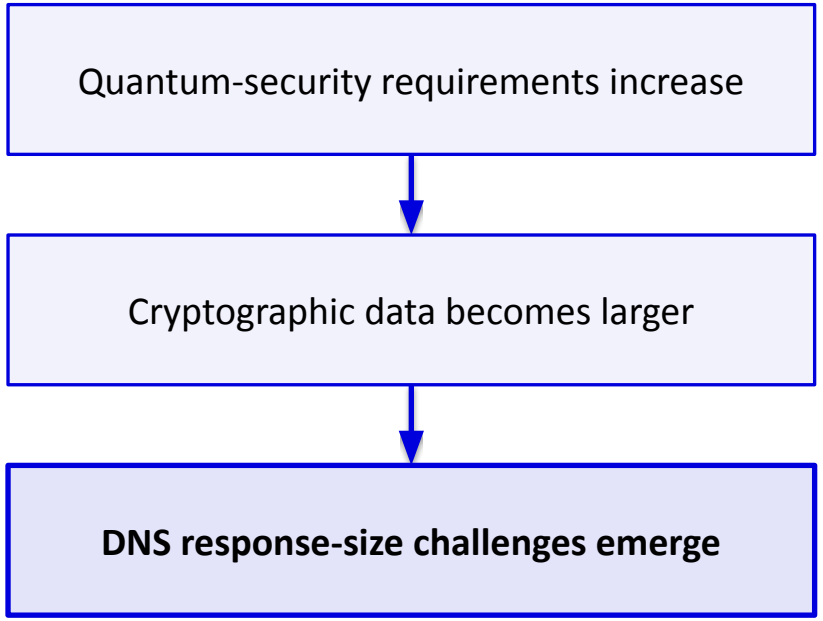
AIORI Hackathon 3

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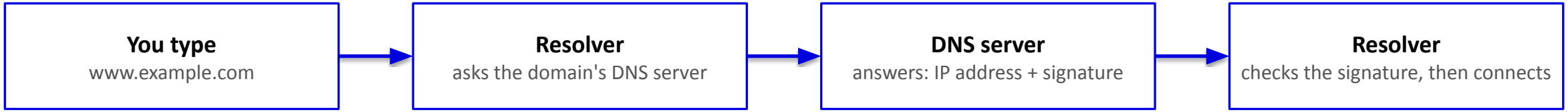
The Problem: DNSSEC in the Quantum Era

What is at risk?

- **DNS** translates domain names into IP addresses so users can reach websites.
- **DNSSEC** uses digital signatures to verify DNS data and protect against tampering [1].
- **Quantum threat:** future quantum computers could break the classical cryptographic algorithms used by DNSSEC (RSA, ECDSA).
- **PQC** (Post-Quantum Cryptography) offers algorithms designed to resist quantum attacks, but some produce much larger keys and signatures [4].



How one lookup works today

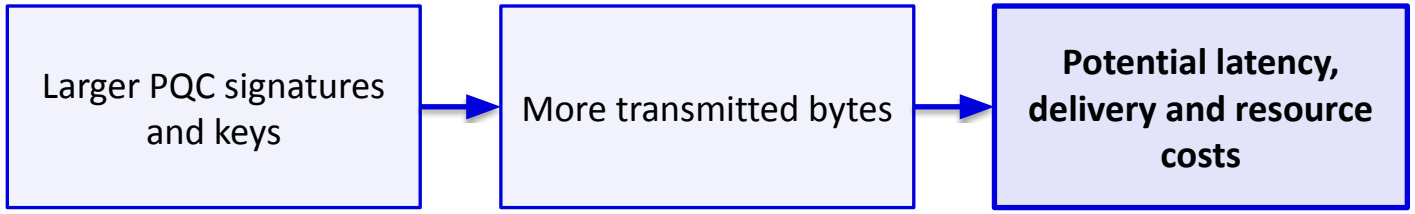


The signature travels inside every answer, so its size is paid on every lookup.

The Technical Challenge

Why does larger cryptographic data matter?

DNS commonly uses **UDP** for responses. Larger DNSSEC responses can exceed the permitted UDP message size (**1,232 bytes** [3, 11]), potentially causing **truncation**, **IP fragmentation**, **packet loss**, or **TCP fallback**.



A truncated answer must be asked for again over TCP: an extra round trip, more bytes and more work for every resolver.

Signature size vs one 1,232-byte UDP packet

Signature type	Size	Fits?
ECDSA P-256 (today)	64 B	yes
RSA-2048 (today)	256 B	yes
SLH-DSA-128s (PQC, we test)	**7,856 B**	**no, 6× too big**

Core problem to investigate

Can quantum-resistant DNSSEC authentication remain practical without excessive transmission and computational overhead?

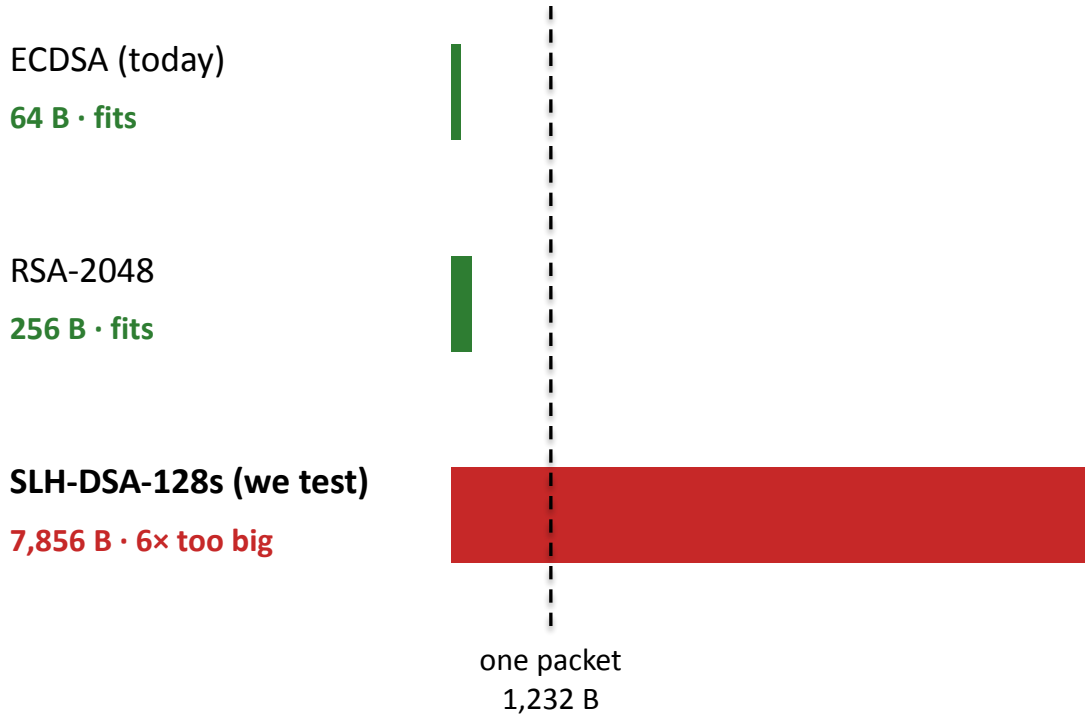
The trade-offs must be measured when comparing MTL and non-MTL workloads:

- **Signing time**
- **Verification time**
- **Transmitted bytes**
- **Cache usage**

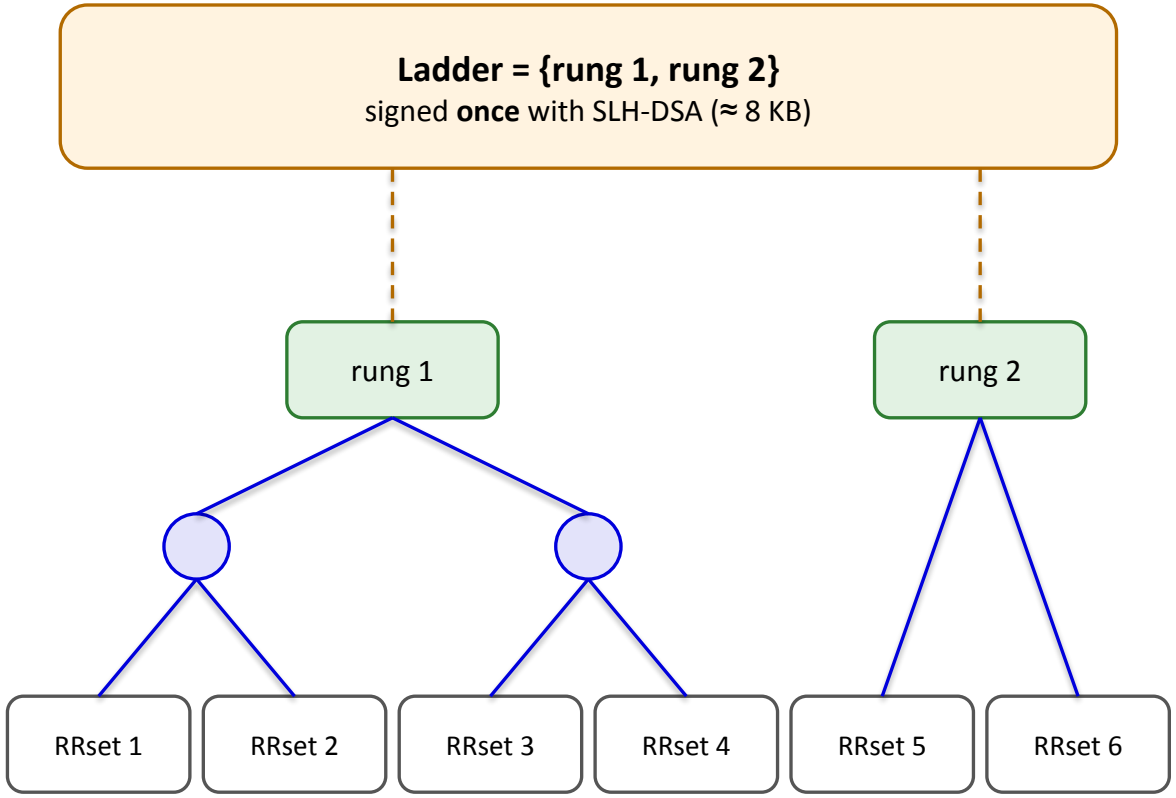
Background (1/3): why size matters

- **DNSSEC** attaches a signature to each set of DNS records, so a resolver can check that an answer is genuine [1].
- Answers normally travel in **one small packet** (≤ 1,232 bytes) [3, 11].
- If an answer is bigger, it is **cut off**, and the resolver must ask again over a slower connection (TCP).
- **SLH-DSA** is the safest quantum-resistant option: it relies only on hash functions and has a tiny key. But its signature is **huge** [4].

Signature size compared with one packet



Background (2/3): how MTL mode works

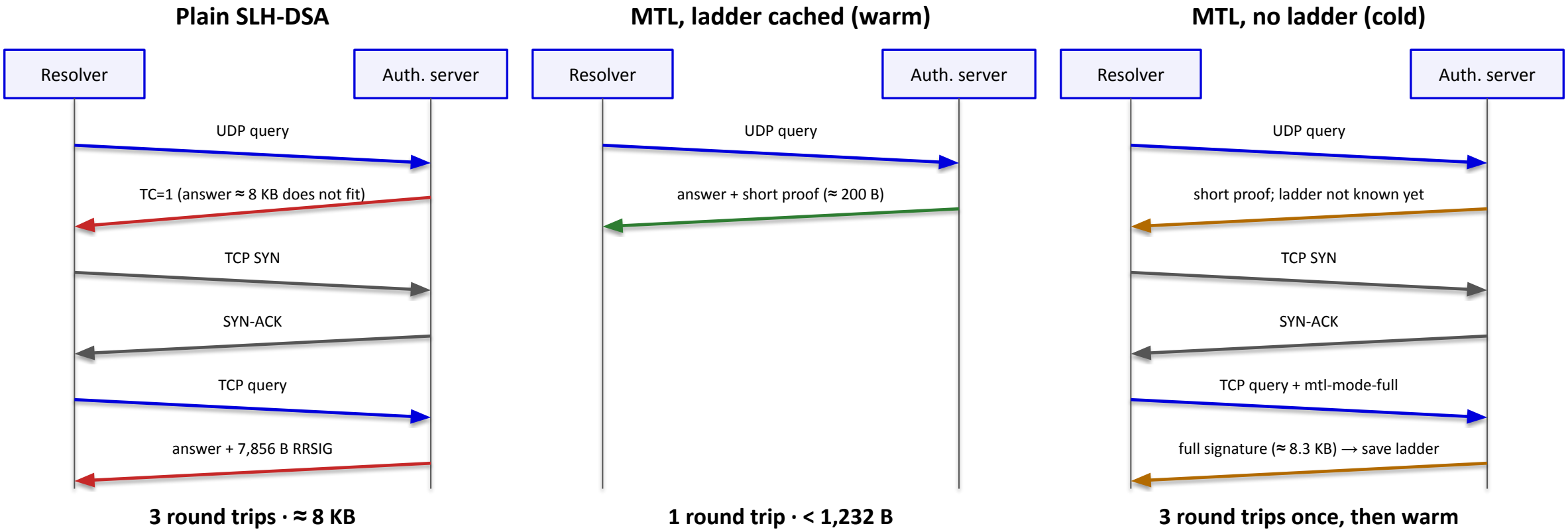


MTL mode is defined in [5]; its use in DNSSEC in [6].

Analogy: instead of a notary signing every page of a long contract, the notary signs the **table of contents** once. Each page carries a short note proving where it sits in that table.

Term	What it means
Ladder	the signed “table of contents” (≈ 8 KB), signed **once**
Short proof	what each record carries: ≈ 200 B instead of 7,856 B
Full signature	short proof + ladder; only needed if the resolver hasn't saved the ladder yet
Node hash	the quick step linking a record to the ladder; far cheaper than a signature

Background (3/3): three ways a lookup can go



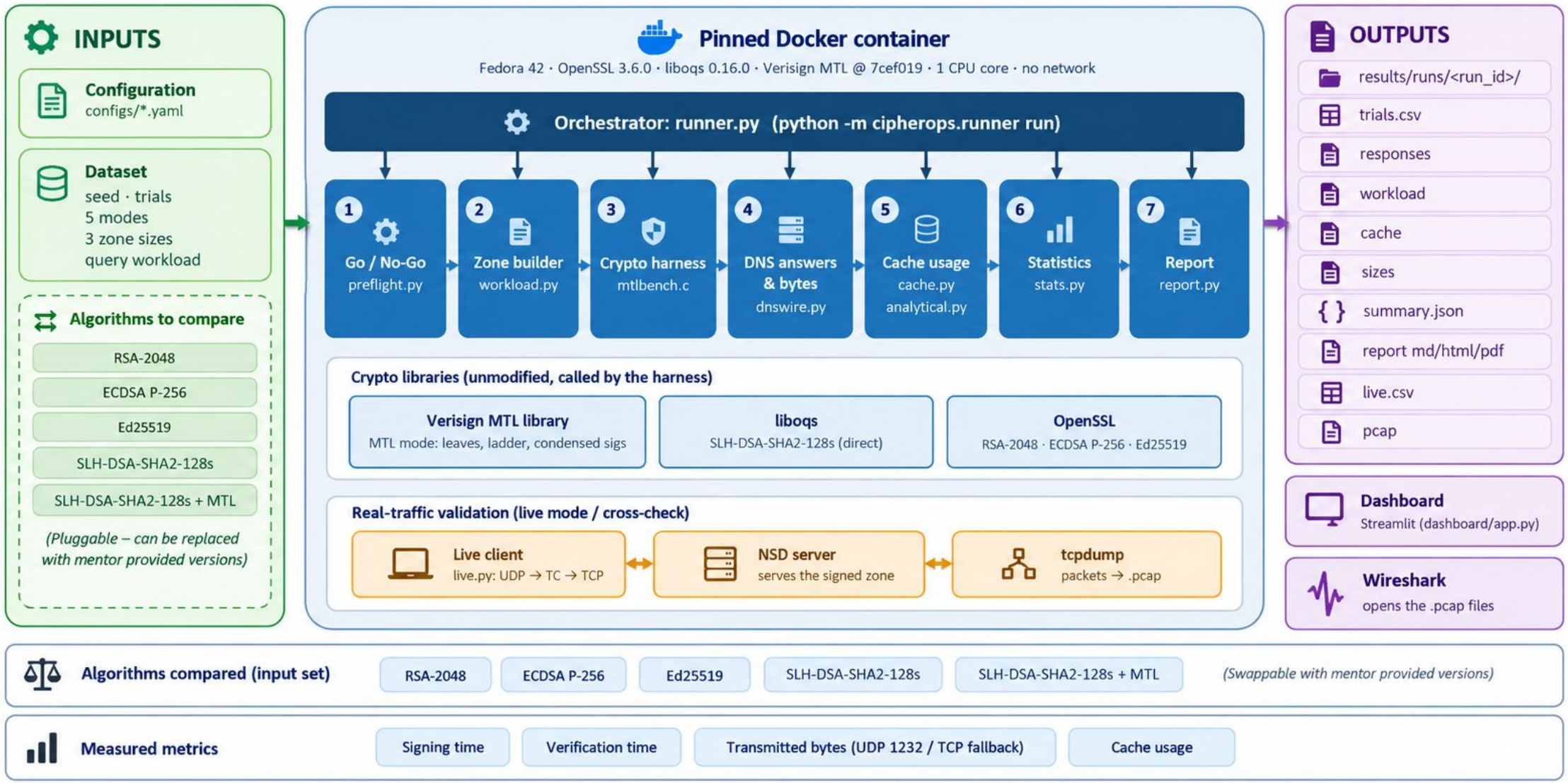
Key point: MTL's benefit depends on **how big the zone is** and **how often the resolver can reuse a saved ladder**. Our test is built around these two factors.

Literature & RFC Review

Only the sources that define what we build or measure.

Ref	Source	What it defines	Used in Cipherops for
[1]	RFC 4033 / 4034 / 4035: DNSSEC	RRSIG and DNSKEY formats; signing input (RFC 4034 §3.1.8.1)	Byte-exact signing input per RRset (workload.py)
[2]	RFC 5155 NSEC3; RFC 9276	Hashed denial of existence; 0 iterations, no salt	NSEC3 chain in every zone; NXDOMAIN answers
[3]	RFC 6891 EDNS(0); RFC 7766	UDP buffer size, TC bit, TCP fallback	1,232 / 4,096-byte limits and TCP retries (dnswire.py)
[4]	NIST FIPS 205: SLH-DSA	The PQC signature scheme (SLH-DSA-SHA2-128s)	The algorithm under test, with and without MTL
[5]	draft-harvey-cfrg-mtl-mode-09	MTL construction; §12 maximum sizes	Analytical size and cache estimates (analytical.py)
[6]	draft-fregly-dnsop-slh-dsa-mtl-dnssec-05	MTL in RRSIG: MTL-Type byte, mtl-mode-full option	Encoding of MTL signatures in DNS answers
[7]	Fregly et al., CT-RSA 2023	Design and size argument for MTL mode	The hypothesis this benchmark tests
[8]	Verisign MTL reference implementation	Supplied library, test suite and CLI tools	MTL signing and verification; go/no-go gate
[9]	liboqs 0.16.0; OpenSSL 3.6.0	Direct SLH-DSA; RSA, ECDSA, Ed25519	Non-MTL and classical baselines
[10]	RFC 5702 / 6605 / 8080	DNSSEC algorithms 8, 13 and 15	Classical reference modes
[11]	DNS Flag Day 2020	1,232-byte EDNS buffer recommendation	Default UDP limit

Proposed Solution



Optimization Proposed

One SLH-DSA signature per zone instead of one per RRset. Large zone (10,005 RRsets), medians of 5 trials:

Metric	SLH-DSA direct (before)	SLH-DSA + MTL (after)	Ed25519 (classical reference)
Signature in each RRSIG	7,856 B	≤ 284 B condensed + one 8,120 B signed ladder per zone	64 B
Sign the whole zone	≈ 92 min (extrapolated)[1]	**0.58 s**	0.37 s
Verify one answer, ladder cached	541 µs	**7.2 µs**[2]	117 µs
Verify the first answer, nothing cached	480 µs	490 µs	103 µs
Signature storage for the zone	78.6 MB	**2.7 MB** (29× less)	0.64 MB
Answers that fit one 1,232-byte UDP packet	0% (median 7,982 B)	**87.4%** (median 411 B)	91.5% (median 190 B)
Bytes on the wire, 762 lookups[3]	11.08 MB; 762 TCP retries	**0.92 MB** (12× less); 246 TCP	0.38 MB; 43 TCP
Time for 762 lookups, 50 ms round trip[3]	125.6 s	**63.5 s** (2× faster)	43.0 s

[1] 1,000 of 10,005 RRsets signed per trial at 0.55 s each, scaled to the zone. Fully measured on the 1,005-RRset zone: 558 s direct vs 0.57 s with MTL. [2] Includes one ladder verification spread over the zone.

[3] Measured live: real client and NSD server, EDNS 1,232 B, 2,000-lookup workload with resolver cache. MTL: 245 truncated answers plus 1 ladder fetch.

Source: not frozen in the repository (see hackathon/results/README.md). The frozen runs are a 3-trial quick demo and a small lab benchmark. This table will be replaced by the full 10-trial run on the reference machine (AWS EC2).

Solution Architecture & Design

UML component and deployment view: two measurement paths, one stack profile, three unmodified crypto libraries.

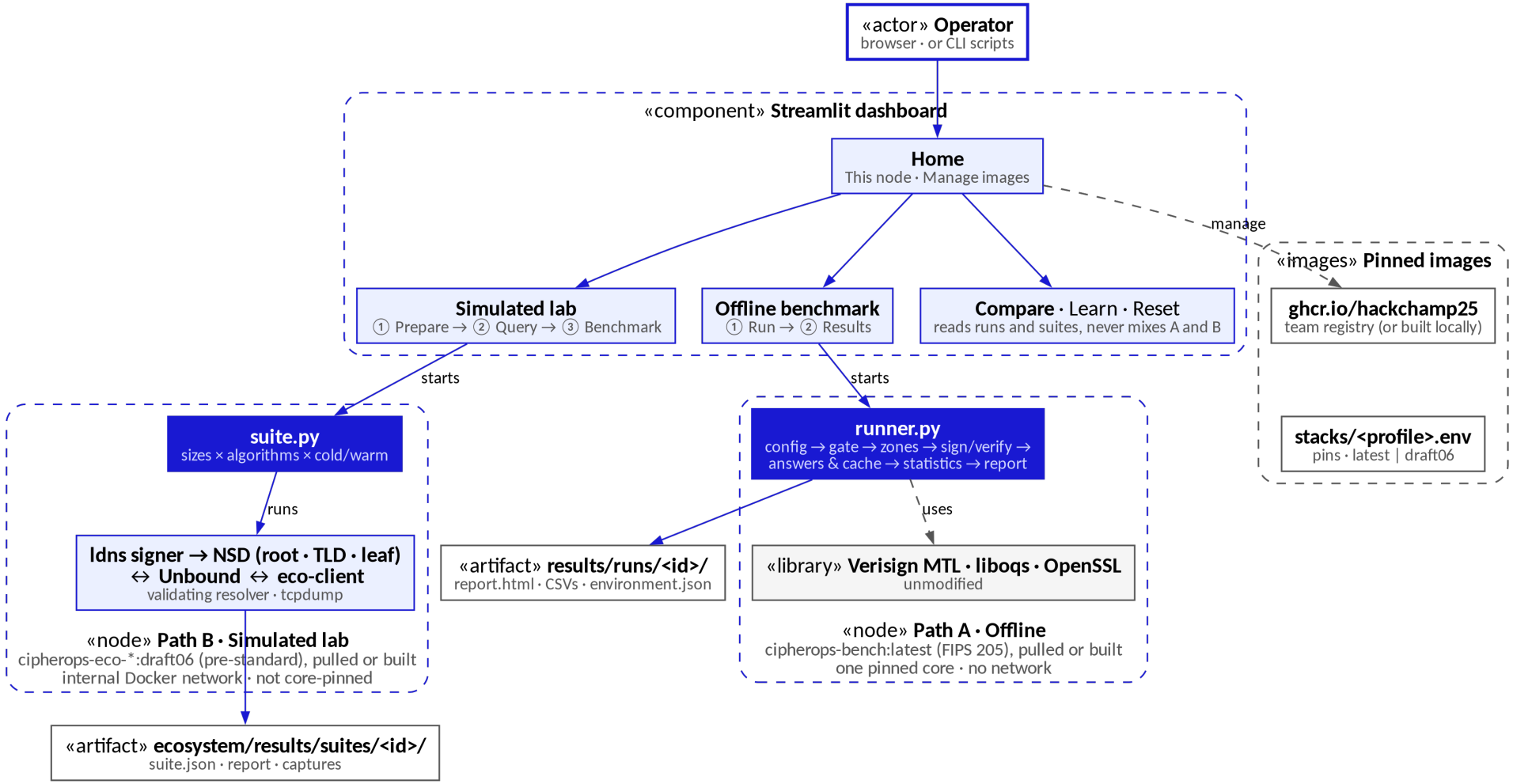
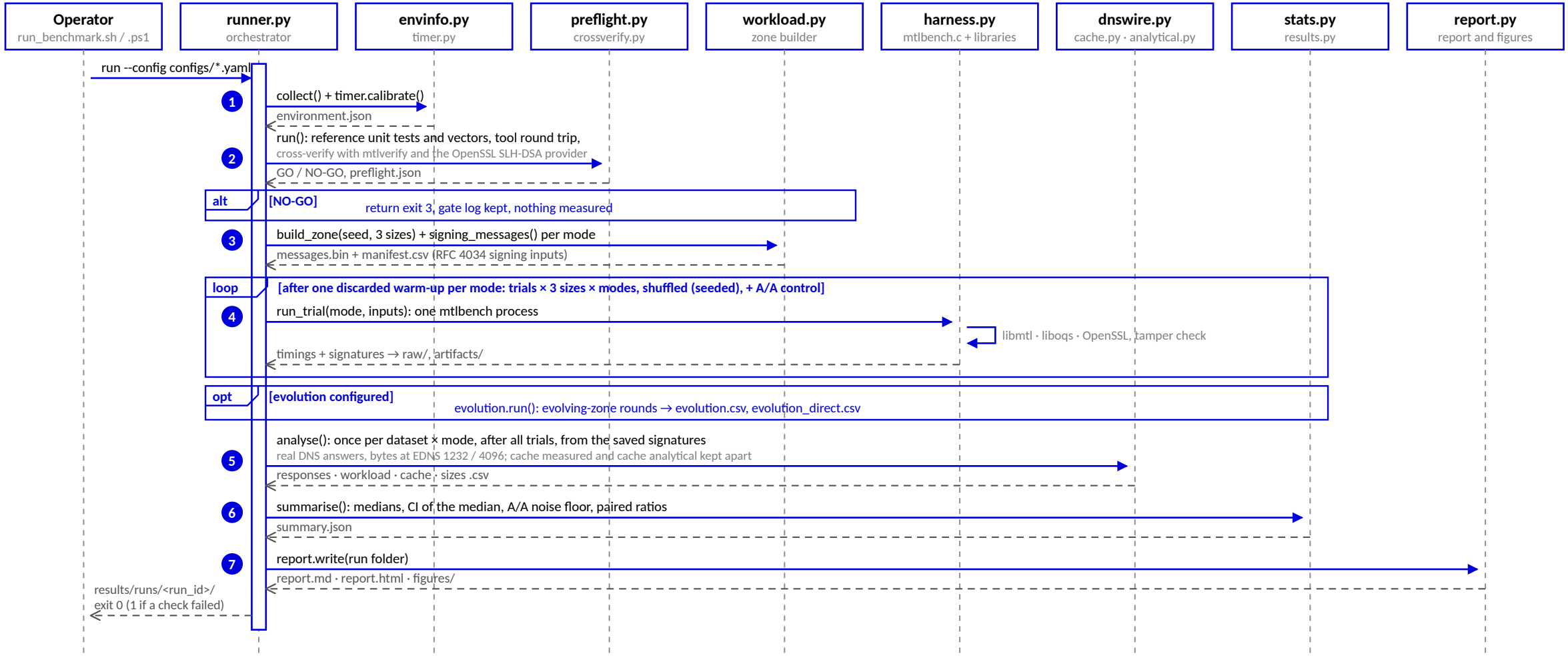


Figure: UML component and deployment view of CipherOps (branch main). The dashboard has two tracks. Offline runs the FIPS 205 build (latest); the lab runs the pre-standard draft06 build, because on latest the lab's resolver answers SERVFAIL for plain SLH-DSA. We report this as a finding: there is no single end-to-end FIPS 205 result.

Runtime Design: One Benchmark Run

UML sequence diagram of runner.py run: the same seven stages the runner prints, from configuration to results.



Later, on a finished run, runner live and runner crosscheck (compose service live) add live.csv, crosscheck.csv and pcap/, then rewrite the report.

Design Approach & Key Decisions

Six design decisions behind the benchmark, and the trade-offs we accept.

1 Reproducible by construction

One pinned Docker image (Fedora 42, OpenSSL

one command run everything.

Evidence: Host needs only Docker and git

2 Fail fast with a Go / No-Go gate

Before any measurement, the supplied MTL library must pass its own tests and a tool round trip. A broken reference stops the run.

Evidence: preflight.py; gate result stored with each run

3 Reference libraries stay untouched

A thin C harness calls Verisign MTL, liboqs and OpenSSL as they are, so timings reflect the reference code and not ours.

Evidence: libraries are called unmodified by mtlbench.c

4 Identical input for every mode

Zones come from one seed. Each RRset's signing input is byte-exact per RFC 4034, so MTL and non-MTL sign the same data.

Evidence: workload.py; seed fixed in the configuration

5 A model, checked on real traffic

dnswire.py and cache.py model UDP limits, TC=1 and TCP retry. A real NSD server with packet capture cross-checks them.

Evidence: 762 lookups measured on a real NSD server

6 Every number is traceable

Each run keeps its configuration, environment record, gate results, raw timings, signatures, pcap and report under one run_id.

Evidence: results/runs/<run_id>/

Trade-offs we accept
One CPU core and no network make runs repeatable, but absolute timings are not production numbers: compare ratios. MTL gains depend on zone size and ladder reuse.

Extension path
Swap in mentor-provided algorithm builds, add ML-DSA-MTL (draft-kaizer-dnsop-ml-dsa-mtl-dnssec), and test multi-core runs and larger zones.

Data Flow Diagram

Level-1 DFD of one benchmark run: from the configuration to the report.

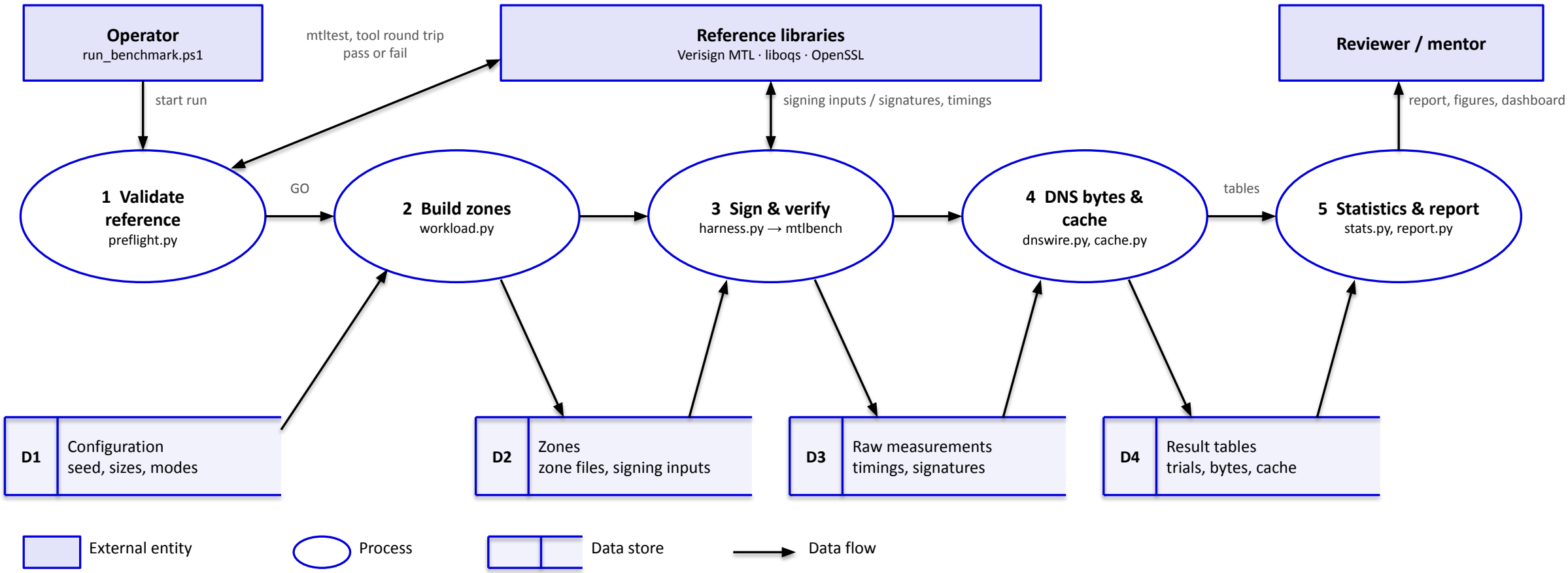


Figure: Data Flow Diagram (DFD)

Deployment Timeline

Milestones, and how the platform will be progressed towards usable deployment.

Sep 30, 2026
Scope and design
PS-036 analysis, RFC and draft review, repository created
Done

Oct 6-10, 2026
Develop MVP prototype-1
Pinned image, go/no-go gate, C harness, zone builder, DNS and cache models, report, dashboard
Done

Oct 10-13, 2026
Quick demo run
3 trials × 3 zones × 5 modes; gate GO; every correctness check passed
Done

Oct 14, 2026
Submission
Report; private GitHub repository transferred to aiori-hackathon; mentor demo
Done

After Oct 16-31
Mentor sessions
Confirming the MTL implementation and incorporating the solution in the code and the live demo.
Planned

Deployment methodology

- **One artifact.** Everything runs from one Docker image built from the repository; the host needs only Docker and git.
- **One command.** A single script builds the image and runs the benchmark (configuration and CPU cores selectable); a second runs the live validation on the finished run.
- **Self-describing runs.** Each run keeps its configuration, environment record, gate results, raw data, signatures, packet captures and report, so every number can be traced and re-run.
- **Tested.** pytest covers workloads, DNS wire format, cache and formulas on the host; the harness and library gates run inside the container.

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[3] J. Damas, M. Graff, P. Vixie. “Extension Mechanisms for DNS (EDNS(0))”. In: *RFC 6891* (2013). J. Dickinson, S. Dickinson, R. Bellis, A. Mankin, D. Wessels. “DNS Transport over TCP – Implementation Requirements”. In: *RFC 7766* (2016).

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[5] J. Harvey et al. “Merkle Tree Ladder (MTL) Mode Signatures”. In: *Internet-Draft draft-harvey-cfrg-mtl-mode-09*, IRTF CFRG (work in progress).

[6] A. Fregly et al. “Stateless Hash-Based Signatures in Merkle Tree Ladder Mode (SLH-DSA-MTL) for DNSSEC”. In: *Internet-Draft draft-fregly-dnsop-slh-dsa-mtl-dnssec-05*, IETF DNSOP (work in progress).

Bibliography II

[7] A. Fregly, J. Harvey, B. S. Kaliski Jr., S. Sheth. “Merkle Tree Ladder Mode: Reducing the Size Impact of NIST PQC Signature Algorithms in Practice”. In: *Topics in Cryptology – CT-RSA 2023*, LNCS 13871, Springer (2023).

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[11] DNS Flag Day. “DNS Flag Day 2020”. dnsflagday.net/2020.

Thank You for Your Attention!